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Blessed G/Michael Catholic Secondary School

1st Semester Mathematics test Two for Grade 11th

Dec, 2018 E.C Time Allowed 1:20'

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I. Choose the best answer among the given alternatives.

D 1. Which one of the following is true about the matrix $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$?

A. It is a square matrix
B. It is a scalar matrix
C. It is an identify matrix
D. All are correct

C 2. Given that $A = \begin{pmatrix} 2 & 1 \\ x & y \end{pmatrix}$ and $\begin{pmatrix} z & w \\ -16 & 8 \end{pmatrix}$ If A = B then what is the value of x

A. 2 B. 1 C. -16 D. 8

A 3. Let $A = \begin{pmatrix} z & 1 \\ x & y \end{pmatrix}$ $B = \begin{pmatrix} z & w \\ -16 & 8 \end{pmatrix}$ and $C = \begin{pmatrix} 4 & 5 \\ 4 & 12 \end{pmatrix}$ then If A+B = C then what is the value of Z?

A. 2 B. 3 C. 4 D. 6

A 4. If $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix}$ what is matrix C If A . B = C?

A. $\begin{pmatrix} 8 & 3 \\ 7 & 7 \end{pmatrix}$ B. $\begin{pmatrix} 6 & 3 \\ 5 & 7 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 8 \\ 5 & 7 \end{pmatrix}$ D. $\begin{pmatrix} 3 & 8 \\ -7 & 7 \end{pmatrix}$

A 5. What is the inverse of matrix $A = \begin{pmatrix} 5 & -4 \\ -3 & 3 \end{pmatrix}$

A. $\begin{pmatrix} 1 & 4/3 \\ 1 & 5/3 \end{pmatrix}$ B. $\begin{pmatrix} -1 & 4/3 \\ 1 & 5/3 \end{pmatrix}$ C. $\begin{pmatrix} 1 & -4/3 \\ 1 & -5/3 \end{pmatrix}$ D. $\begin{pmatrix} -1 & 4/3 \\ -1 & 5/3 \end{pmatrix}$

C 6. Given that A = (aij) such that a ij = j – i and A is a 2x2 matrix then find matrix A?

A. $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ B. $\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$ C. $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ D. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

b C 7. Which one of the following is false about $\begin{pmatrix} 0 & 4 & -6 \\ -4 & 0 & 2 \\ 6 & -2 & 0 \end{pmatrix}$

A. It is skew symmetric matrix
B. It is a symmetric matrix
C. $A^t = A$
D. $A + A^t = 0$

8. Which one of the following is true about invertible matrix?

A. $\begin{pmatrix}1\\A\end{pmatrix}t=(A^t)^{-1}$ B. $\begin{pmatrix}ka & kb\\ c & d\end{pmatrix}=k\begin{pmatrix}a & b\\ c & d\end{pmatrix}$, C. $(AB)^{-1}=A^{-1}B^{-1}$ D. all are correct

9. If $A=\begin{pmatrix}2 & 1\\ 5 & 3\end{pmatrix}$ then $A^t=$ ----------

A. $\begin{pmatrix}2 & 5\\ 1 & 3\end{pmatrix}$ B. $\begin{pmatrix}5 & 2\\ 3 & 1\end{pmatrix}$ C. $\begin{pmatrix}3 & 5\\ 1 & 2\end{pmatrix}$ D. $\begin{pmatrix}1 & 2\\ 3 & 5\end{pmatrix}$

10. Which one of the following is an identity matrix

A. $\begin{pmatrix}0 & 2\\ 2 & 0\end{pmatrix}$ B. $\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}$ C. $\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ D. $\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$

11. Which one of the following is true about the equation? $\begin{cases}2x-y=5\\ 6x-3y=15\end{cases}$

A. It has unique solution.
B. It has no solution.
C. It has infinite solution.
D. It is impossible to determine

12. Which one of the following is true for matrices A & B.

A. $(A^{-1})^t=A^tA$ B. $(AB)^t=A^tB^t$, C. $(AB)^{-1}=B^{-1}A^{-1}$ D. $A^{-1}=A^n$

II. Short Answer (write the answer only)

1. Let $A=\begin{pmatrix}1 & 2\\ -2 & 1\end{pmatrix}$ and $B=\begin{pmatrix}-3 & 11\\ 1 & -21\end{pmatrix}$ then find AB ---------- $\begin{pmatrix}-1 & -31\\ -7 & -43\end{pmatrix}$

2. Reduce the matrix $\begin{pmatrix}2 & 1 & 3\\ 1 & 4 & 2\\ 0 & 5 & 1\end{pmatrix}$ to RREF. ---------- $\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}$

3. Solve the equation $\begin{cases}4x+2y+3z=6\\ 7y+2x=3z\\ -3x-9y+13=-2z\end{cases}$ $x=$ ---- $y=$ ---- $z=$ ---- $\mathbb{R}$ $\mathbb{R}$ $\mathbb{R}$ → Infinite sol=

4. If $\begin{pmatrix}1 & 1 & 3\\ 0 & 0 & 3-k/2\end{pmatrix}$ is the RREF of system of equation the for what value of k, the system of equation has
   a. no solution ---------- $k\neq 6$
   b. Infinite solution ---------- $k=6$
   c. one solution ---------- Can't have

   but we don't have the solution side or else the question is not augmented form.

5. If $A=(a_{ij})_{3\times n}$ matrix and $B=(b_{ij})_{n\times m}$ then for what values of n and m AB and BA are defined
   $m=$ ---------- & $n=$ ----------